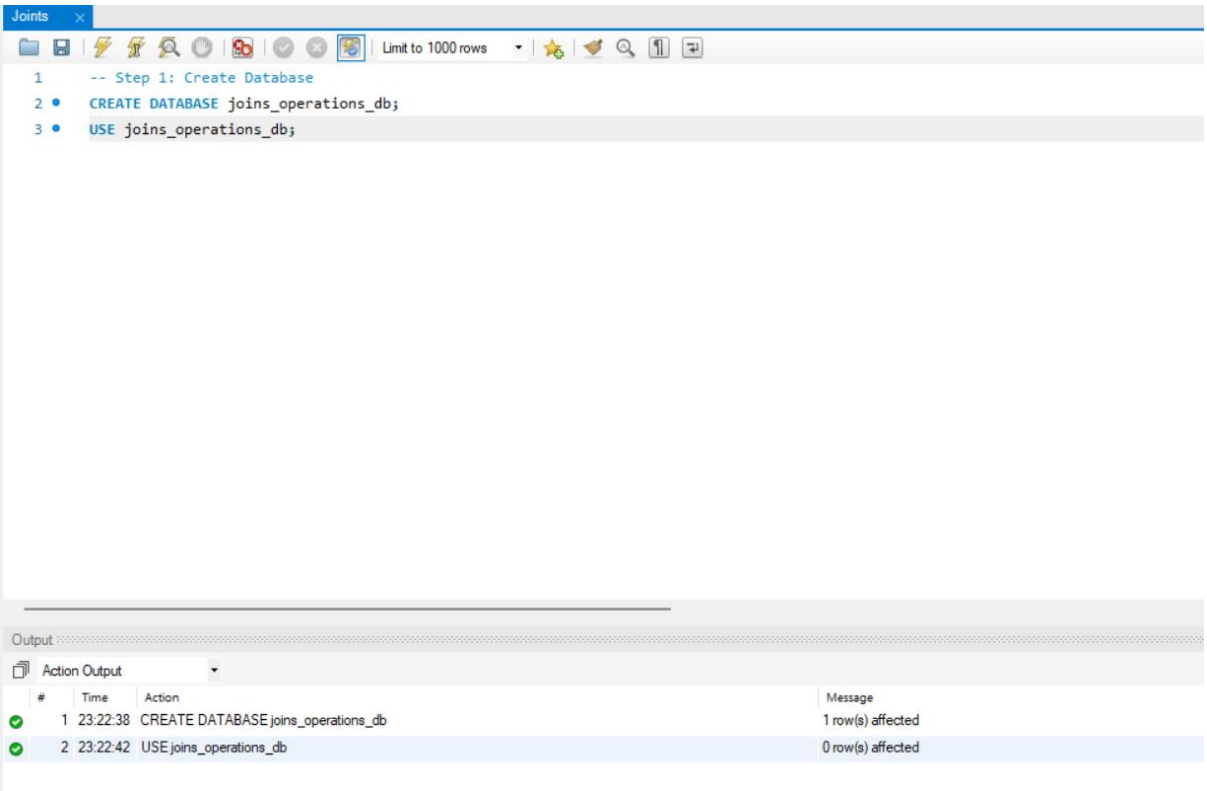
JOINS

Ili.Harshika (2520030607)

Question :

Perform SQL JOIN and SET operations on student and address data using CROSS JOIN, INNER JOIN, NATURAL JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, UNION, UNION ALL, INTERSECT and MINUS/EXCEPT, including filtering of unmatched records and displaying address availability using CASE

Step 1 : Create Database



Joints

```sql
-- Step 1: Create Database
CREATE DATABASE joins_operations_db;
USE joins_operations_db;
-- Step 2: Create Tables
CREATE TABLE class(
    id INT,
    name VARCHAR(30)
);

CREATE TABLE class_info(
    id INT,
    address VARCHAR(30)
);
```

Output

Action Output

#	Time	Action	Message
1	23:22:38	CREATE DATABASE joins_operations_db	1 row(s) affected
2	23:22:42	USE joins_operations_db	0 row(s) affected
3	23:23:15	CREATE TABLE class(id INT, name VARCHAR(30))	0 row(s) affected
4	23:23:18	CREATE TABLE class_info(id INT, address VARCHAR(30))	0 row(s) affected

Joints

```sql
);

CREATE TABLE class_info(
    id INT,
    address VARCHAR(30)
);
-- Step 3: Insert Data
INSERT INTO class VALUES
(1,'Harshika'),
(2,'Pranitha'),
(3,'Angela'),
(4,'Bhanu Priya'),
(5,'Nikshitha');

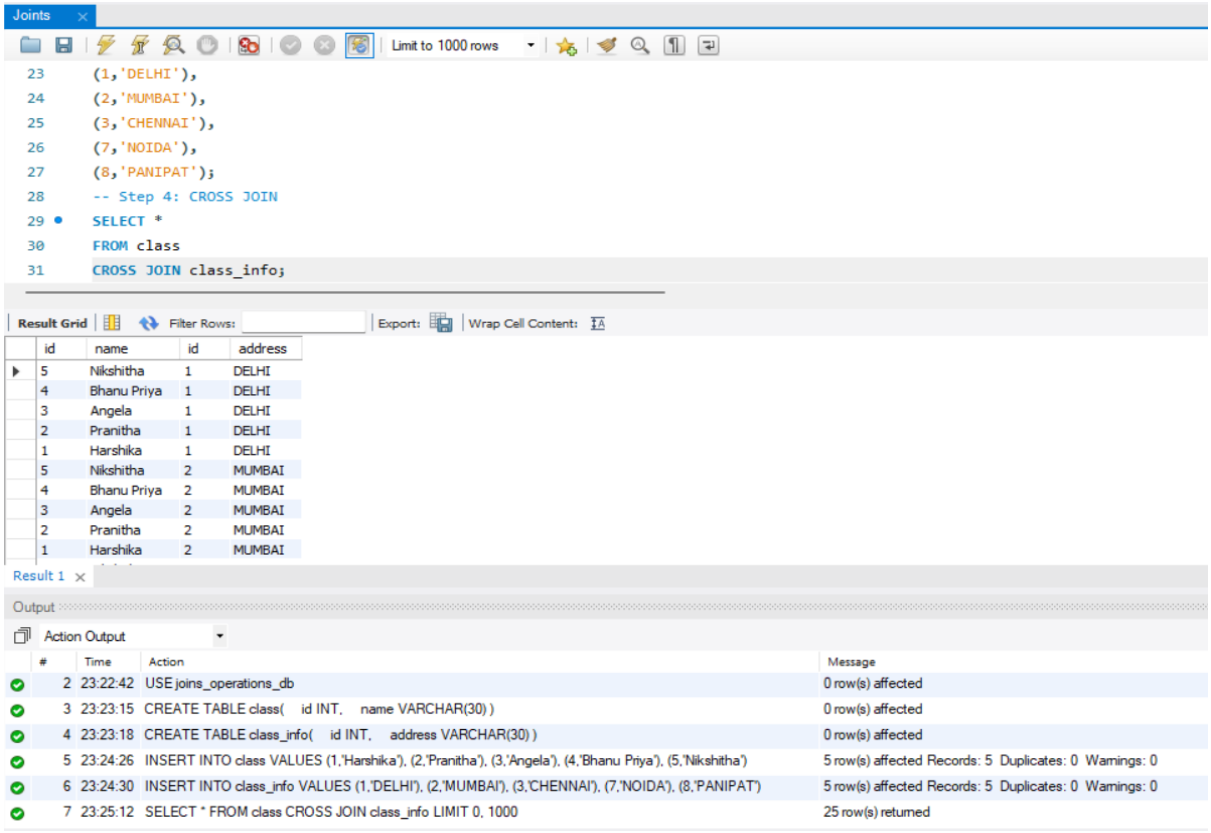
INSERT INTO class_info VALUES
(1,'DELHI'),
(2,'MUMBAI'),
(3,'CHENNAI'),
(7,'NOIDA'),
(8,'PANIPAT');
```

Output

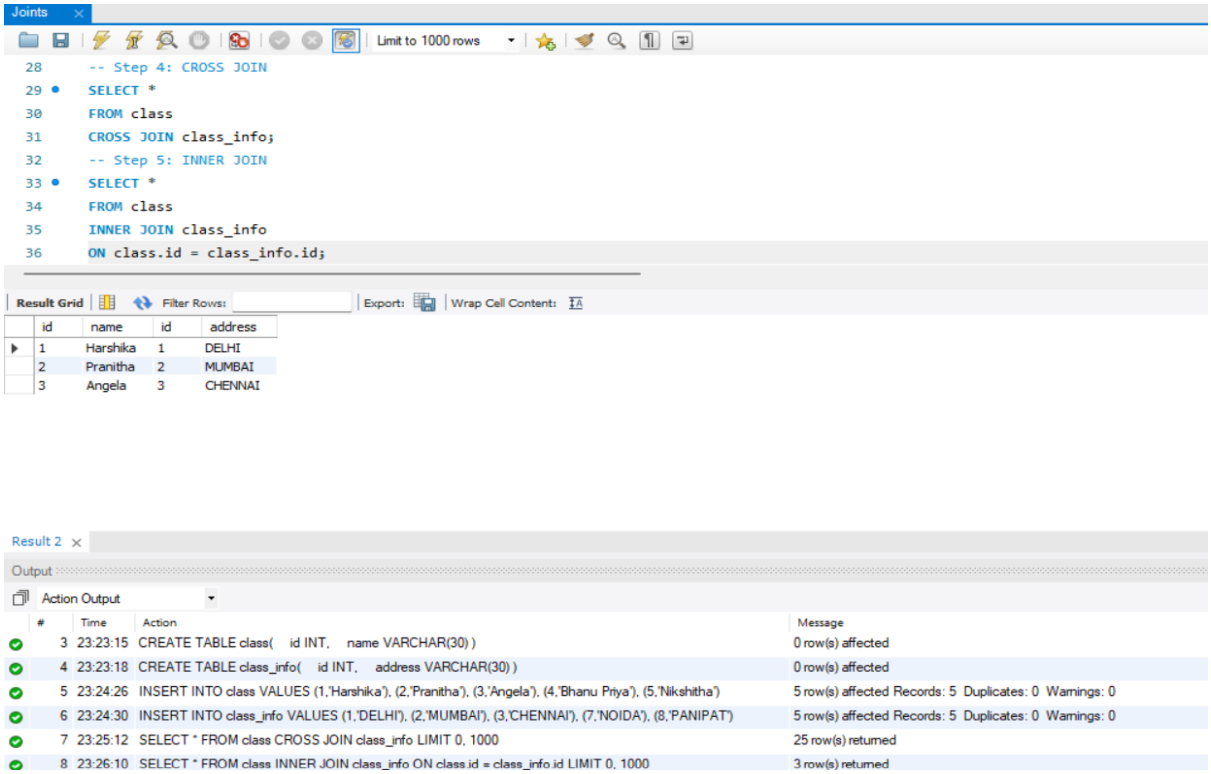
Action Output

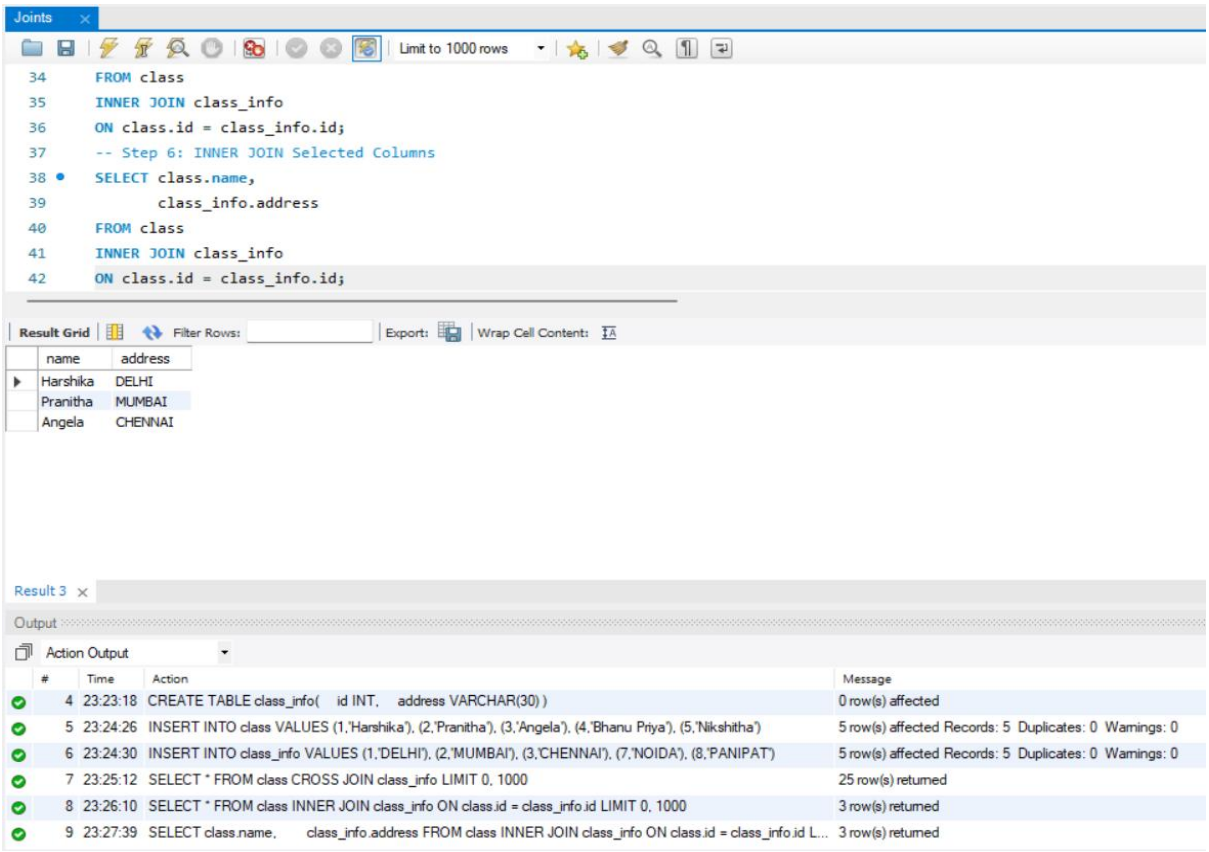
#	Time	Action	Message
1	23:22:38	CREATE DATABASE joins_operations_db	1 row(s) affected
2	23:22:42	USE joins_operations_db	0 row(s) affected
3	23:23:15	CREATE TABLE class(id INT, name VARCHAR(30))	0 row(s) affected
4	23:23:18	CREATE TABLE class_info(id INT, address VARCHAR(30))	0 row(s) affected
5	23:24:26	INSERT INTO class VALUES (1,'Harshika'), (2,'Pranitha'), (3,'Angela'), (4,'Bhanu Priya'), (5,'Nikshitha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	23:24:30	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI'), (7,'NOIDA'), (8,'PANIPAT')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

Step 2 : CROSS JOIN

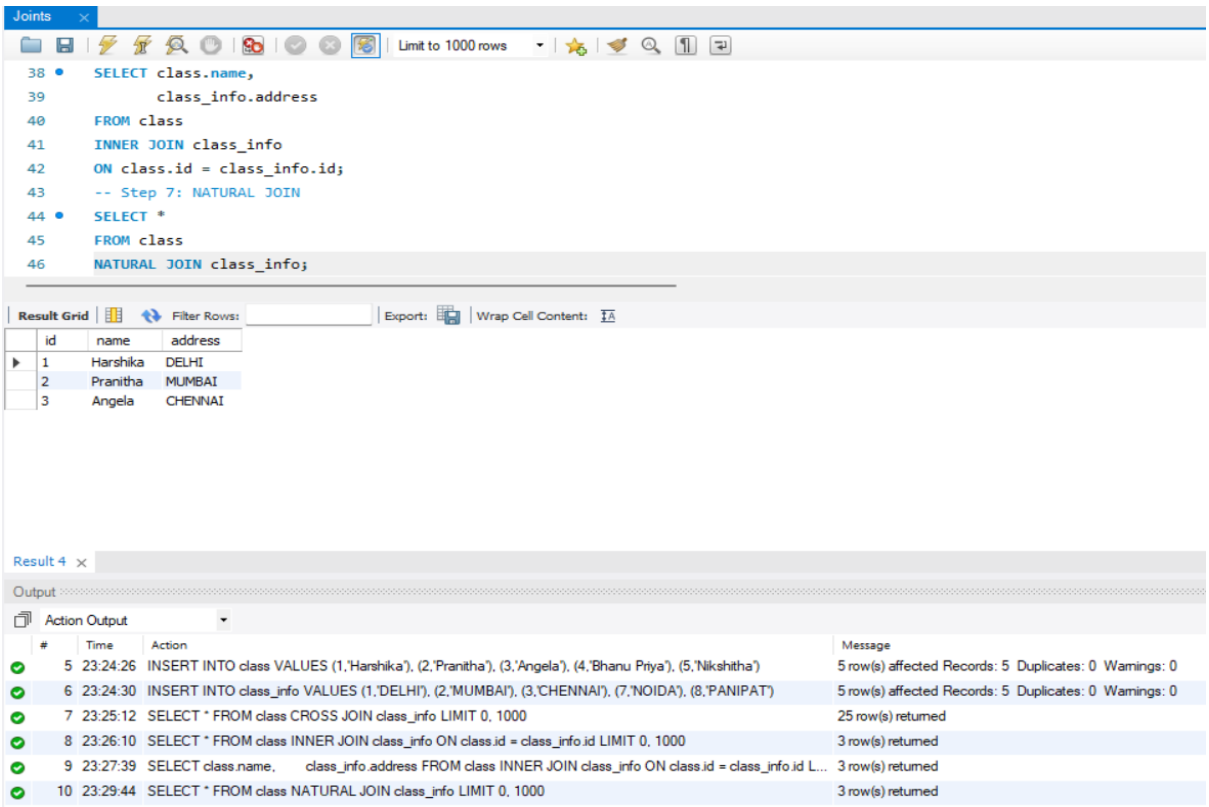


Step 3 : INNER JOIN

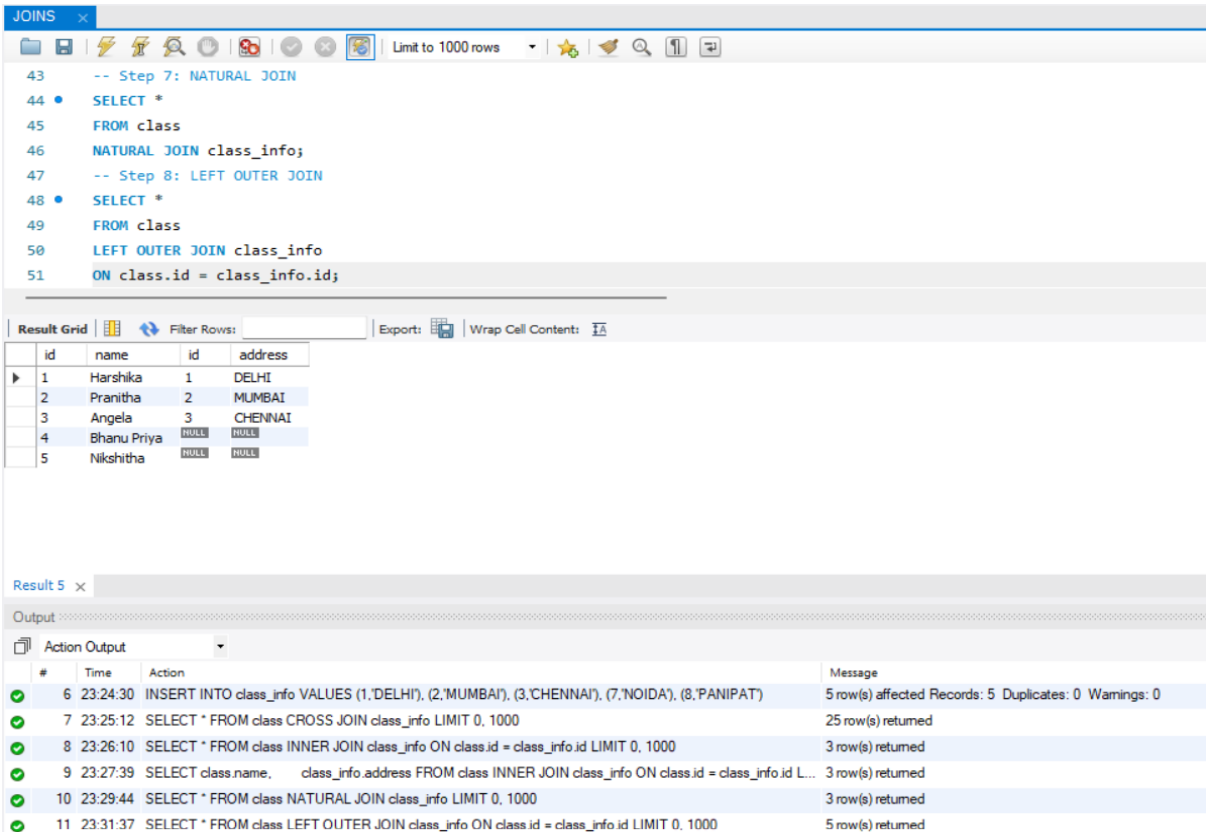


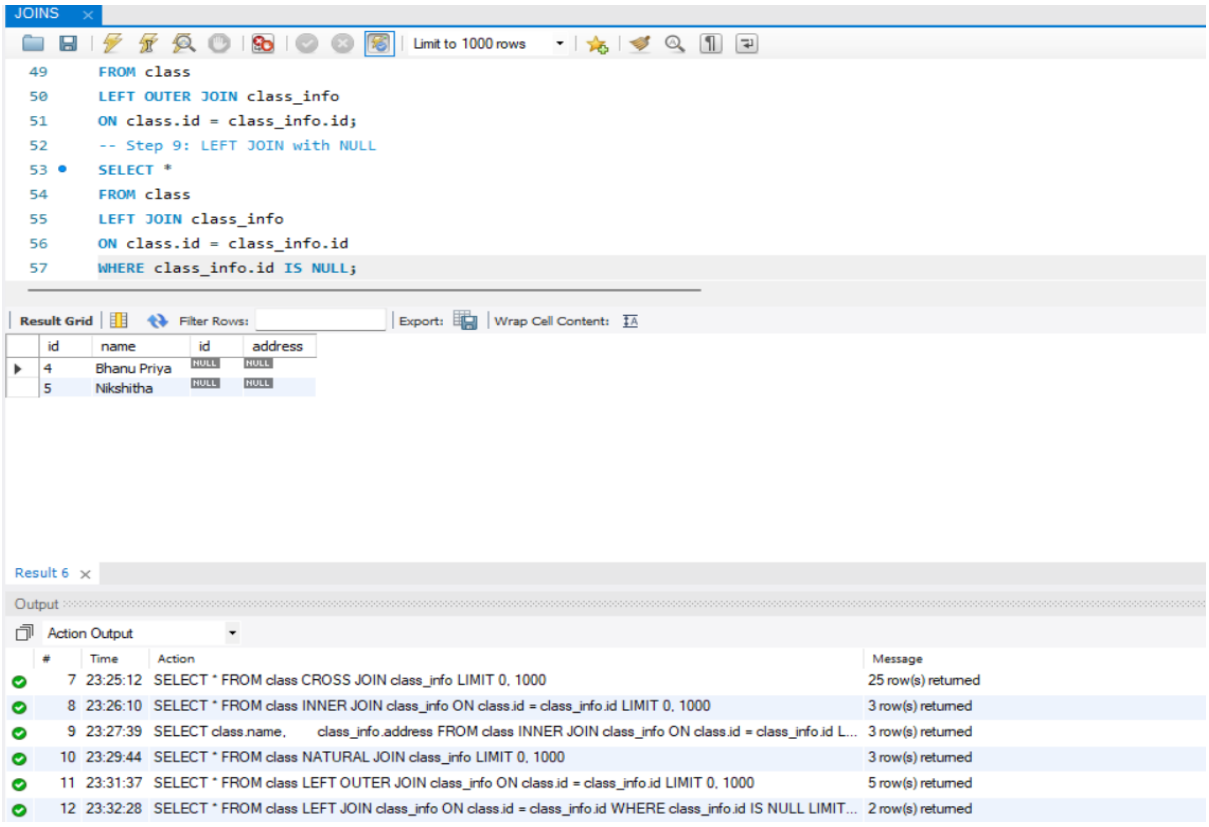


Step 4 : NATURAL JOIN



Step 5 : LEFT OUTER JOIN





Step 6 : RIGHT OUTER JOIN

JOINS

```
54      FROM class
55      LEFT JOIN class_info
56      ON class.id = class_info.id
57      WHERE class_info.id IS NULL;
58      -- Step 10: RIGHT OUTER JOIN
59  •   SELECT *
60      FROM class
61      RIGHT OUTER JOIN class_info
62      ON class.id = class_info.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	id	address
1	Harshika	1	DELHI
2	Pranitha	2	MUMBAI
3	Angela	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Result 7

Output

Action Output

#	Time	Action	Message
8	23:26:10	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s) returned
9	23:27:39	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id L...	3 row(s) returned
10	23:29:44	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
11	23:31:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
12	23:32:28	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
13	23:33:36	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned

JOINS

```
60      FROM class
61      RIGHT OUTER JOIN class_info
62      ON class.id = class_info.id;
63      -- Step 11: RIGHT JOIN with NULL
64  •   SELECT *
65      FROM class
66      RIGHT JOIN class_info
67      ON class.id = class_info.id
68      WHERE class.id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	id	address
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Result 8

Output

Action Output

#	Time	Action	Message
9	23:27:39	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id L...	3 row(s) returned
10	23:29:44	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
11	23:31:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
12	23:32:28	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
13	23:33:36	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
14	23:34:21	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned

Step 7: FULL OUTER JOIN

JOINS

```
72    LEFT JOIN class_info
73    ON class.id = class_info.id
74
75    UNION
76
77    SELECT *
78    FROM class
79    RIGHT JOIN class_info
80    ON class.id = class_info.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	id	address
1	Harshika	1	DELHI
2	Pranitha	2	MUMBAI
3	Angela	3	CHENNAI
4	Bhanu Priya	NULL	NULL
5	Nikshitha	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Result 9

Output

Action Output

#	Time	Action	Message
10	23:29:44	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
11	23:31:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
12	23:32:28	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
13	23:33:36	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
14	23:34:21	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
15	23:35:40	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIG...	7 row(s) returned

JOINS

```
86    WHERE class_info.id IS NULL
87
88    UNION
89
90    SELECT *
91    FROM class
92    RIGHT JOIN class_info
93    ON class.id = class_info.id
94    WHERE class.id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	id	address
4	Bhanu Priya	NULL	NULL
5	Nikshitha	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Result 10

Output

Action Output

#	Time	Action	Message
11	23:31:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
12	23:32:28	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
13	23:33:36	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
14	23:34:21	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
15	23:35:40	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIG...	7 row(s) returned
16	23:36:15	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNI...	4 row(s) returned

Step 8 : SET OPERATIONS

JOINS

```
85      ON class.id = class_info.id
86      WHERE class_info.id IS NULL
87
88      UNION
89
90      SELECT *
91      FROM class
92      RIGHT JOIN class_info
93      ON class.id = class_info.id
94      WHERE class.id IS NULL;
95      -- Step 14: Create Set Operation Tables
96      CREATE TABLE first_table(
97          id INT,
98          name VARCHAR(30)
99      );
100
101     CREATE TABLE second_table(
102         id INT,
103         name VARCHAR(30)
104     );
```

Output

Action Output

#	Time	Action	Message
13	23:33:36	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
14	23:34:21	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
15	23:35:40	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIG...	7 row(s) returned
16	23:36:15	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNI...	4 row(s) returned
17	23:37:08	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
18	23:37:11	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected

JOINS

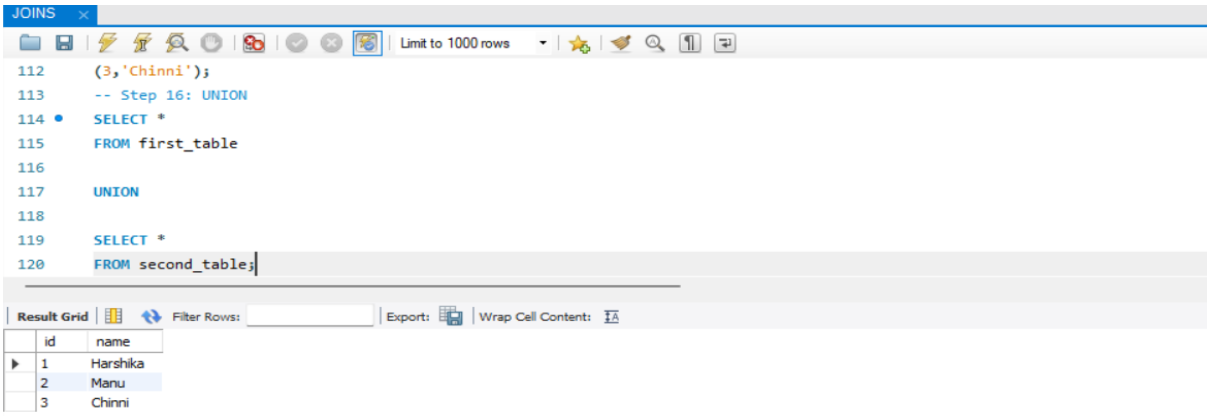
```
93      ON class.id = class_info.id
94      WHERE class.id IS NULL;
95      -- Step 14: Create Set Operation Tables
96      CREATE TABLE first_table(
97          id INT,
98          name VARCHAR(30)
99      );
100
101     CREATE TABLE second_table(
102         id INT,
103         name VARCHAR(30)
104     );
105     -- Step 15: Insert Set Operation Data
106     INSERT INTO first_table VALUES
107     (1,'Harshika'),
108     (2,'Manu');
109
110     INSERT INTO second_table VALUES
111     (2,'Manu'),
112     (3,'Chinni');
```

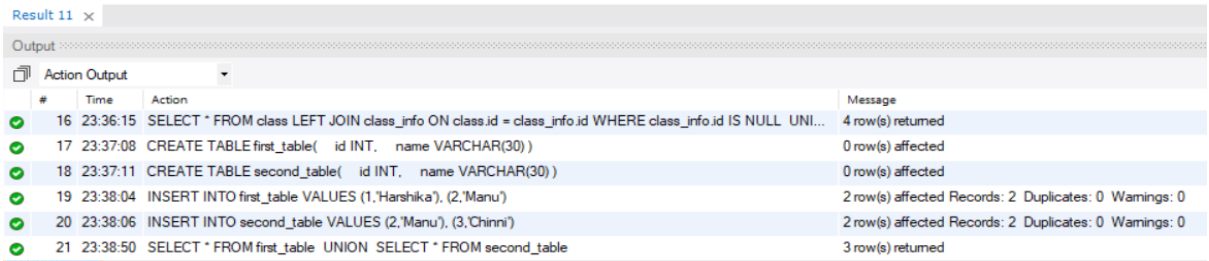
Output

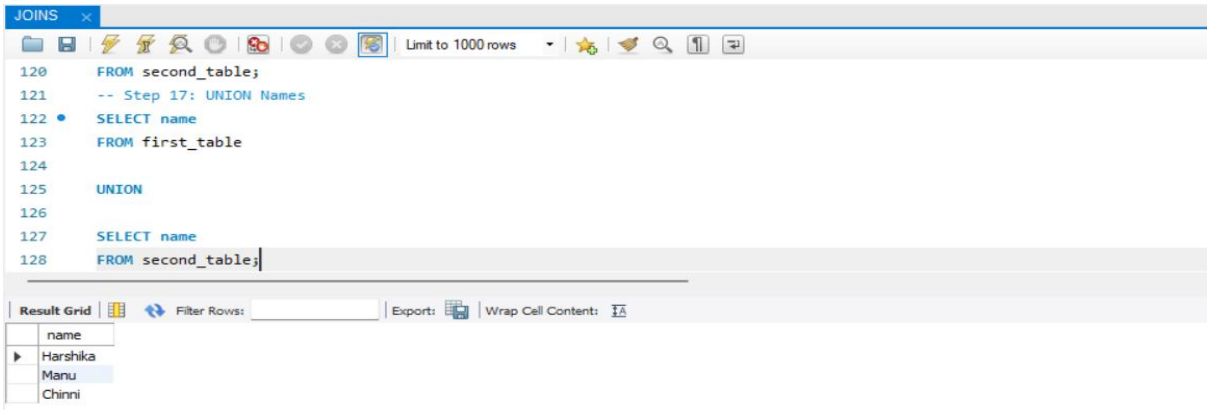
Action Output

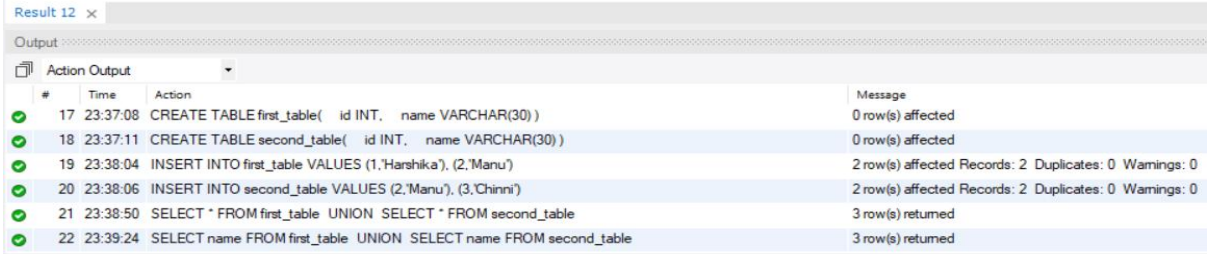
#	Time	Action	Message
15	23:35:40	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIG...	7 row(s) returned
16	23:36:15	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNI...	4 row(s) returned
17	23:37:08	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
18	23:37:11	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
19	23:38:04	INSERT INTO first_table VALUES (1,'Harshika'), (2,'Manu')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20	23:38:06	INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

➢ UNION


JOINS
Limit to 1000 rows
(3,'Chinni');
-- Step 16: UNION
SELECT *
FROM first_table
UNION
SELECT *
FROM second_table;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
id name
1 Harshika
2 Manu
3 Chinni


Result 11
Output
Action Output
Time Action Message
16 23:36:15 SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNI... 4 row(s) returned
17 23:37:08 CREATE TABLE first_table(id INT, name VARCHAR(30)) 0 row(s) affected
18 23:37:11 CREATE TABLE second_table(id INT, name VARCHAR(30)) 0 row(s) affected
19 23:38:04 INSERT INTO first_table VALUES (1,'Harshika'), (2,'Manu') 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20 23:38:06 INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni') 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21 23:38:50 SELECT * FROM first_table UNION SELECT * FROM second_table 3 row(s) returned


JOINS
Limit to 1000 rows
FROM second_table;
-- Step 17: UNION Names
SELECT name
FROM first_table
UNION
SELECT name
FROM second_table;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
name
Harshika
Manu
Chinni


Result 12
Output
Action Output
Time Action Message
17 23:37:08 CREATE TABLE first_table(id INT, name VARCHAR(30)) 0 row(s) affected
18 23:37:11 CREATE TABLE second_table(id INT, name VARCHAR(30)) 0 row(s) affected
19 23:38:04 INSERT INTO first_table VALUES (1,'Harshika'), (2,'Manu') 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20 23:38:06 INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni') 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21 23:38:50 SELECT * FROM first_table UNION SELECT * FROM second_table 3 row(s) returned
22 23:39:24 SELECT name FROM first_table UNION SELECT name FROM second_table 3 row(s) returned

➢ UNION ALL

JOINS

```
128     FROM second_table;
129     -- Step 18: UNION ALL
130 •   SELECT *
131     FROM first_table
132
133     UNION ALL
134
135     SELECT *
136     FROM second_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name
1	Harshika
2	Manu
2	Manu
3	Chinni

Result 13

Output

Action Output

#	Time	Action	Message
18	23:37:11	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
19	23:38:04	INSERT INTO first_table VALUES (1,'Harshika'), (2,'Manu')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20	23:38:06	INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21	23:38:50	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
22	23:39:24	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
23	23:40:03	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned

JOINS

```
137     -- Step 19: Count UNION ALL Records
138 •   SELECT COUNT(*)
139     FROM (
140         SELECT * FROM first_table
141
142         UNION ALL
143
144         SELECT * FROM second_table
145     ) AS A;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)
4

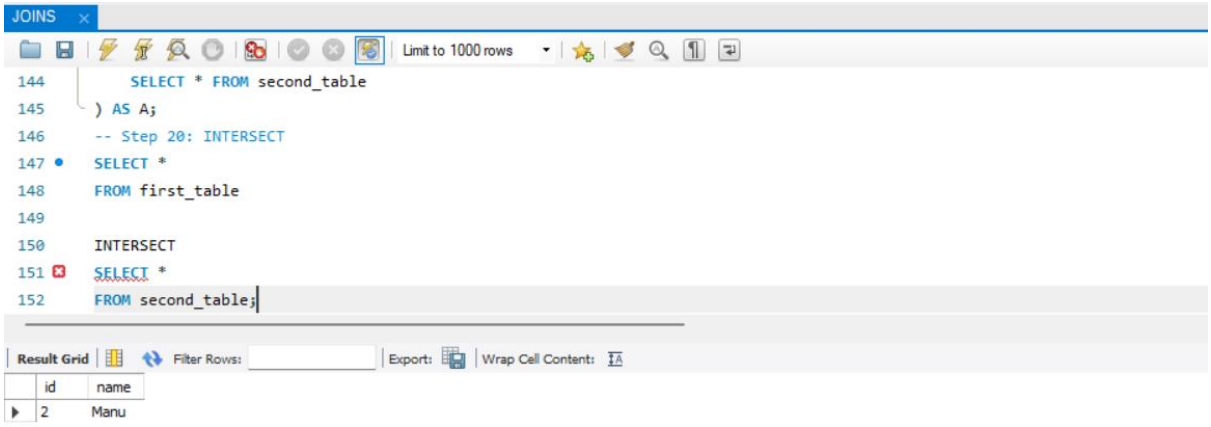
Result 14

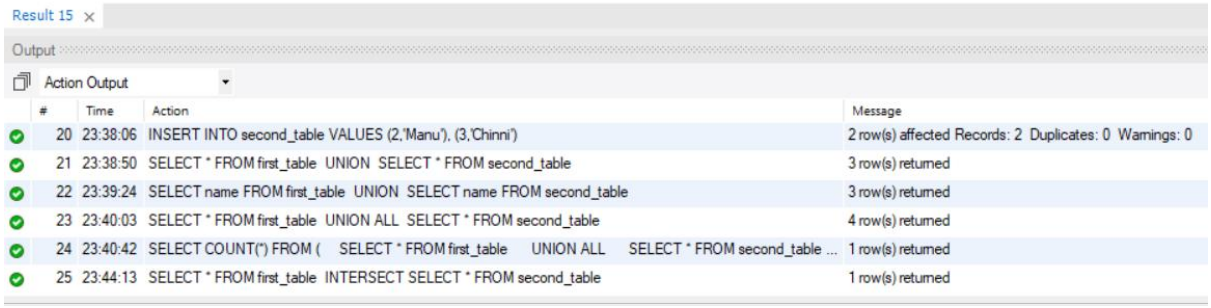
Output

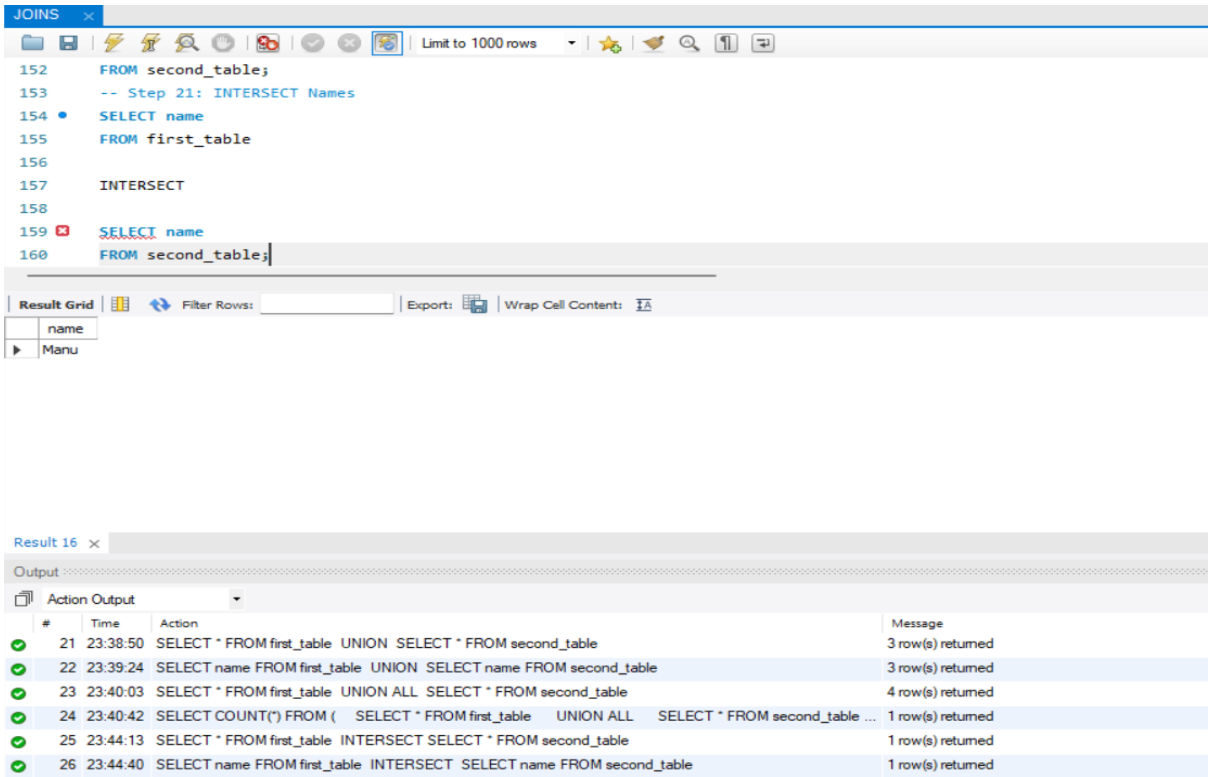
Action Output

#	Time	Action	Message
19	23:38:04	INSERT INTO first_table VALUES (1,'Harshika'), (2,'Manu')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20	23:38:06	INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21	23:38:50	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
22	23:39:24	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
23	23:40:03	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
24	23:40:42	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned

➢ INTERSECT


JOINS
Limit to 1000 rows
144 SELECT * FROM second_table
145) AS A;
146 -- Step 20: INTERSECT
147 SELECT *
148 FROM first_table
149
150 INTERSECT
151 SELECT *
152 FROM second_table;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
id name
2 Manu


Result 15
Output
Action Output
Time Action Message
20 23:38:06 INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni') 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21 23:38:50 SELECT * FROM first_table UNION SELECT * FROM second_table 3 row(s) returned
22 23:39:24 SELECT name FROM first_table UNION SELECT name FROM second_table 3 row(s) returned
23 23:40:03 SELECT * FROM first_table UNION ALL SELECT * FROM second_table 4 row(s) returned
24 23:40:42 SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ... 1 row(s) returned
25 23:44:13 SELECT * FROM first_table INTERSECT SELECT * FROM second_table 1 row(s) returned


JOINS
Limit to 1000 rows
152 FROM second_table;
153 -- Step 21: INTERSECT Names
154 SELECT name
155 FROM first_table
156
157 INTERSECT
158
159 SELECT name
160 FROM second_table;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
name
Manu
Result 16
Output
Action Output
Time Action Message
21 23:38:50 SELECT * FROM first_table UNION SELECT * FROM second_table 3 row(s) returned
22 23:39:24 SELECT name FROM first_table UNION SELECT name FROM second_table 3 row(s) returned
23 23:40:03 SELECT * FROM first_table UNION ALL SELECT * FROM second_table 4 row(s) returned
24 23:40:42 SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ... 1 row(s) returned
25 23:44:13 SELECT * FROM first_table INTERSECT SELECT * FROM second_table 1 row(s) returned
26 23:44:40 SELECT name FROM first_table INTERSECT SELECT name FROM second_table 1 row(s) returned

➢ MINUS

JOINS

Limit to 1000 rows

```
161    -- Step 22: MINUS
162    SELECT f.*
163    FROM first_table f
164    WHERE NOT EXISTS (
165        SELECT 1
166        FROM second_table s
167        WHERE s.id = f.id
168        AND s.name = f.name
169    );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name
1	Harshika

first_table 17

Output

Action Output

#	Time	Action	Message
22	23:39:24	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
23	23:40:03	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
24	23:40:42	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned
25	23:44:13	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
26	23:44:40	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned
27	23:45:24	SELECT f.* FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WHERE s.id...	1 row(s) returned

JOINS

Limit to 1000 rows

```
169    );
170    -- Step 23: MINUS Names
171    SELECT f.name
172    FROM first_table f
173    WHERE NOT EXISTS (
174        SELECT 1
175        FROM second_table s
176        WHERE s.name = f.name
177    );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name
Harshika

first_table 18

Output

Action Output

#	Time	Action	Message
23	23:40:03	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
24	23:40:42	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned
25	23:44:13	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
26	23:44:40	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned
27	23:45:24	SELECT f.* FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WHERE s.id...	1 row(s) returned
28	23:45:58	SELECT f.name FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WHER...	1 row(s) returned